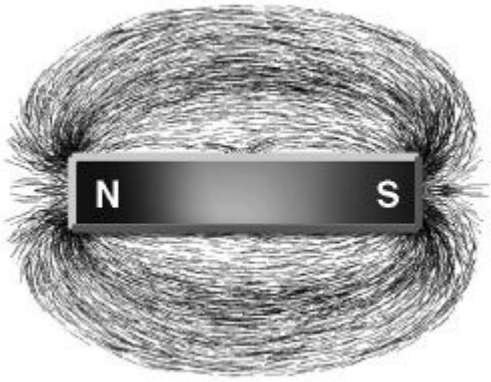


Figure 3.1 Iron filings sprinkled near a bar magnet trace out its magnetic field.

It is the pervasive, space-filling capability of magnetic fields that makes them so useful. An airport metal detector's magnetic field seeps through your clothes and causes metallic objects to give off their own magnetic fields—fields that then exert an influence back on the detector, causing its alarm to sound. An MRI's magnetic field seeps into your body, causing particular atoms to gyrate in just the right way to generate their own magnetic fields—fields that the machine can detect and decode into a picture of internal tissues. The earth's magnetic field seeps through the compass casing and turns the needle, causing it to point along an arc that, as a result of eons-long geophysical processes, is aligned in a nearly south–north direction.

Magnetic fields are one familiar kind of field, but Faraday also analyzed another: the *electric field.* This is the field that causes your wool scarf to crackle, zaps your hand in a carpeted room when you touch a metal doorknob, and makes your skin tingle when you're up in the mountains during a powerful lightning storm. And if you happened to examine a compass during such a storm, the way its magnetic needle deflected this way and that as the bolts of electric lightning flashed nearby would have given you a hint of a deep interconnection between electric and magnetic fields—something first discovered by the Danish physicist Hans Oersted and investigated thoroughly by Faraday through painstaking experimentation. Just as developments in the stock market can affect the bond market which can then affect the stock market, and so on, these scientists found that changes in an electric field can produce changes in a nearby magnetic field, which can then cause changes in the electric field, and so on. Maxwell found the mathematical underpinnings